

A European **call option** is written on a non-dividend-paying stock.

The **current stock price (S_0) is ₹100**, the **strike price (K) is ₹105**, the **risk-free interest rate (r) is 6% per annum (continuous)**, the **volatility (σ) is 20%**, and the **time to maturity (T) is 1 year**.

ANSWERS

- d_1 : 0.1560
- d_2 : -0.0440
- Value of the Call Option: 8.49
- Option Delta: 0.56
- Payoff at Maturity (Stock Price = 120): 15
- Profit at Maturity: 6.51
- Option Status at Expiry: In-the-money